

2026 DEFRA, Local Council and Urban Observatory Sensor Data Analysis

Table of contents

1 Summary	2
2 Maps	3
3 Monthly data	3
3.1 PM2.5 comparisons between close sensors	3
4 Weekly data	7
4.1 PM2.5 comparisons between close sensors	7
4.2 NOx and NO2 comparisons between same sensors	11
5 Attributes affecting PM2.5 concentraions	16
5.1 Wind direction and speed	16
5.2 Altitude	18
5.3 Temperature	18

Following a meeting with council member, Michael Terry, we were made aware of NCC's restrictions in only using calibrated DEFRA-approved and -owned AURN instrumentation and locally-managed automatic monitoring sensors. The Urban Observatory (UO) has a collection of precision MONITOR sensors and MESH sensors. The UO MESH sensors are not calibrated and not approved by DEFRA and therefore NCC cannot use them. We need to verify that the UO MONITOR sensor data agrees with that of DEFRA and AURN sensors. We begin by comparing the data from UO MESH and MONITOR sensors against the DEFRA and NCC sensors. Any and all analysis has been conducted on 2026 data.

For the purposes of this report we will use the following acronyms:

- DEFRA-owned and approved AURN sensors (DEFRA)
- Locally managed automatic monitoring sensors (Local)

- UO precision MONITOR sensors (UO-Mon)
- UO MESH sensors (UO-Mesh)

1 Summary

We have highlighted the following as areas for priority action/attention:

- For PM2.5 data, the UO-Mon and UO-Mesh sensors have low agreement. We will proceed with using the UO-Mon sensors only.
- Some UO-Mesh and UO-Mon sensors have faulty wind direction and speed data, shown as constant over the course of a day, week and month, we have taken some decommissioned sensor data from 2025 to produce analysis.
- DEFRA Newcastle Centre sensor is systematically rounding PM2.5 values to nearest integer, this will be brought to NCC's attention.
- To validate the UO-Mon sensors we compared them to their closest DEFRA/Local sensors, all UO-Mon sensors compared had very good agreement with their closest DEFRA/Local sensor. Some UO-Mon sensors were compared with further (in distance) DEFRA/Local sensors in order to compare all DEFRA/Local sensors. We found increase in distance to change the relationship between sensors to resemble an exponential. This was not fully consistent and we would appreciate some expert UO opinion on this matter.

The report is structured as follows:

- Long time interval analysis, March 2026, comparing PM2.5 concentrations from closest precision sensors (DEFRA/Local/UO-Mon) to UO-Mesh sensors and closest DEFRA/Local sensors to UO-Mon sensors. March 2026 reasonably demonstrates a longer time period where these sensors could differ in 15min - 1hr intervalled collection. We chose month that was relatively recent with minimal missing sensors and values. Caveat: there is a short time period between 19th-21st where data is missing from all sensors.
- Short time interval analysis, 30th March - 5th April 2026, comparing PM2.5 concentrations from closest UO-Mon sensors to UO-Mesh sensors and NOx and NO2 concentrations from same sensors for DEFRA, Local and UO-Mon sensors. We chose a shorter time scale for this analysis as we wanted to discern if the UO-Mesh sensors showed any correlation and temporal similarities (although temporal analysis has not been included in this report) to their closest UO-Mon sensors. If any of the UO-Mesh sensors had shown correlation to the UO-Mon sensors, we could still use them for a shorter time period indicator system. NOx and NO2 concentrations degrade very fast, so only a short time scale is appropriate for their analysis case.
- Extraneous factors affecting PM2.5 concentrations were also analysed over long time scale, March 2025. First we examine wind effects, comparing speed and direction to concentration in 2026 was not possible as stated in the above reasoning. With 2025 data, we take old, decommissioned sensors and perform similar analysis with closest UO-Mon

sensors and Local sensors. DEFRA sensors were not considered as they are situated further away. We are continuing to compare other factors such as temperature and elevation.

2 Maps

The three maps in Figure 1 show closest precision sensors (DEFRA/Local/UO-Mon) to UO-Mesh sensors, closest DEFRA/Local sensors to UO-Mon sensors, and closest UO-Mon sensors to UO-Mesh sensors. These pairs will be used to analyse correlations between closest sensors, validate UO-Mon sensors and understand if UO-Mesh sensors can be used in practice. The distances (km and birds flight) between these pairs are stated in Section 3 correlational analysis.

3 Monthly data

We took March 2026 to look at correlation comparisons between the closest precision (DEFRA/Local/UO-Mon) to UO-Mesh sensors and closest DEFRA/Local sensors to UO-Mon sensors. March was chosen as the dataset had reliable consistent data, i.e. minimal missing values and missing sensor data.

3.1 PM2.5 comparisons between close sensors

Closest precision (DEFRA/Local/UO-Mon) to UO-Mesh sensors, Figure 2 below shows scatter plots of these relationships. These show little to no reliability of the UO mesh sensors.

Figure 3 shows closest UO-Mon sensors to DEFRA/Local sensors, scatter plots and time series, where only the closest UO-Mon sensor is included for each DEFRA/Local sensor. This hopes to validate the UO-Mon sensor data and from both sets of plots we see very good agreement between the sensors. Figure 4 shows the same comparison as Figure 3 however the UO-Mon sensors under consideration (UO-Mon Gateshead Tyne Bridge and UO-Mon RVI) have already been compared to other DEFRA/Local sensors in Figure 3 that are much closer, we see worse agreement for two of the sensor comparisons here which is expected due to the distance. A summary table of correlational information and distances between sensors is given in Table 1 from which we see high correlation between all close sensors. We use both Pearson's and Spearman's correlation coefficients. Pearson's evaluates linear relationships and proportional change between variables while Spearman's monotonic relationships which discerns if variables are changing together, not necessarily at a constant rate as Pearson's would let us assume. It is important to remember that other relationship (non-linear) might also exist as we see in the scaling differences between Loal A1 Dunston sensor and Local Gateshead Lychgate Court sensors with their closest UO-Mon sensor. This could be attributed to their further distance



Figure 1: Maps of (left to right) closest precision sensors (DEFRA/Local/UO-Mon) to UO-Mesh sensors, closest DEFRA/Local sensors to UO-Mon sensors, and closest UO-Mon sensors to UO-Mesh sensors.

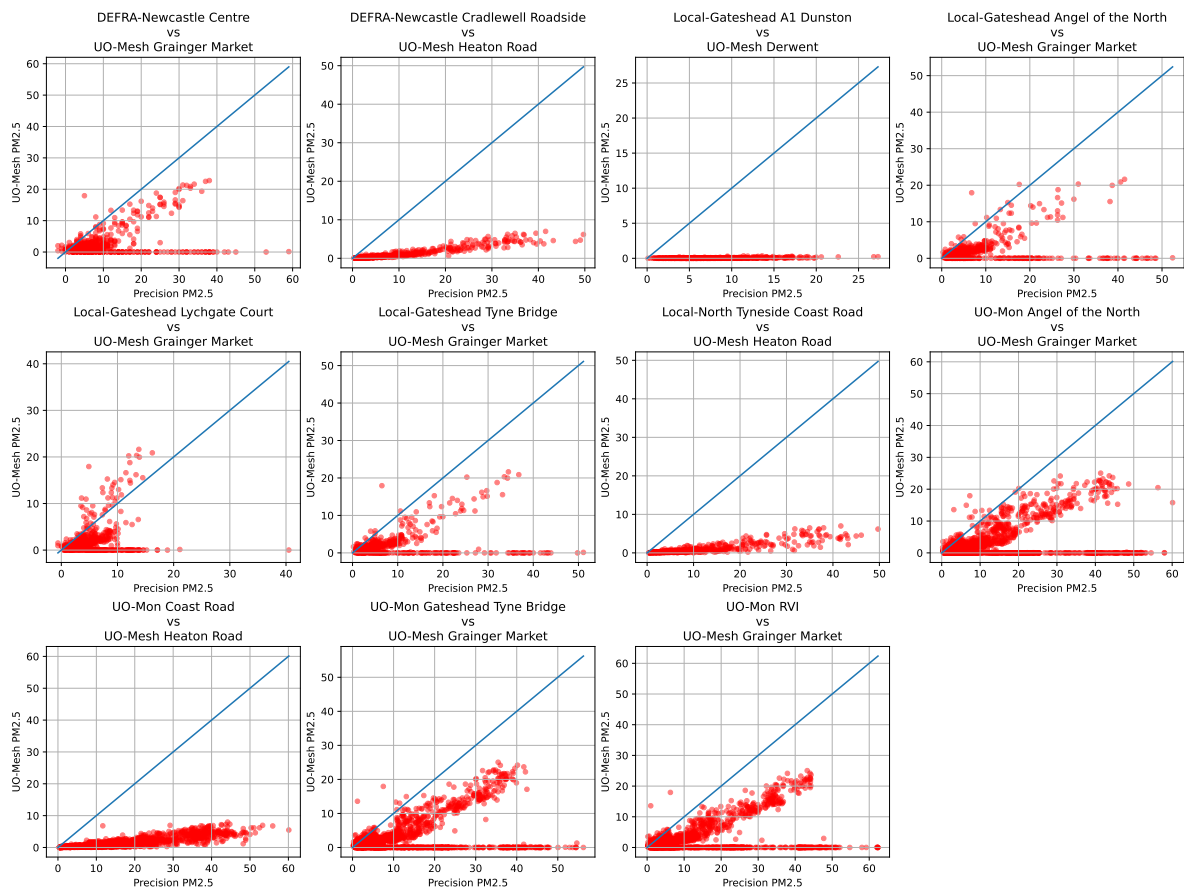


Figure 2: PM2.5 concentrations comparisons between UO-Mesh and closest DEFRA/Local/UE-Mon sensors over long time interval.

from thier closest UO-Mon if it were not for DEFRA Newcastle Cradlewell Roadside case which is further away from it's closest UO-Mon than the Local Gateshead Lychgate Court sensor. It could also be attributed to the fact that we are comparing the same monitor sensor for 4 of the DEFRA/Local sensors as there are no closer UO-Mon sensors in the area. Another issue is the case of DEFRA Newcastle Centre sensor, this is systematically rounding values to the closest integer, we will see this more clearly when analysing weekly data but here you can spot it in the time series plot.

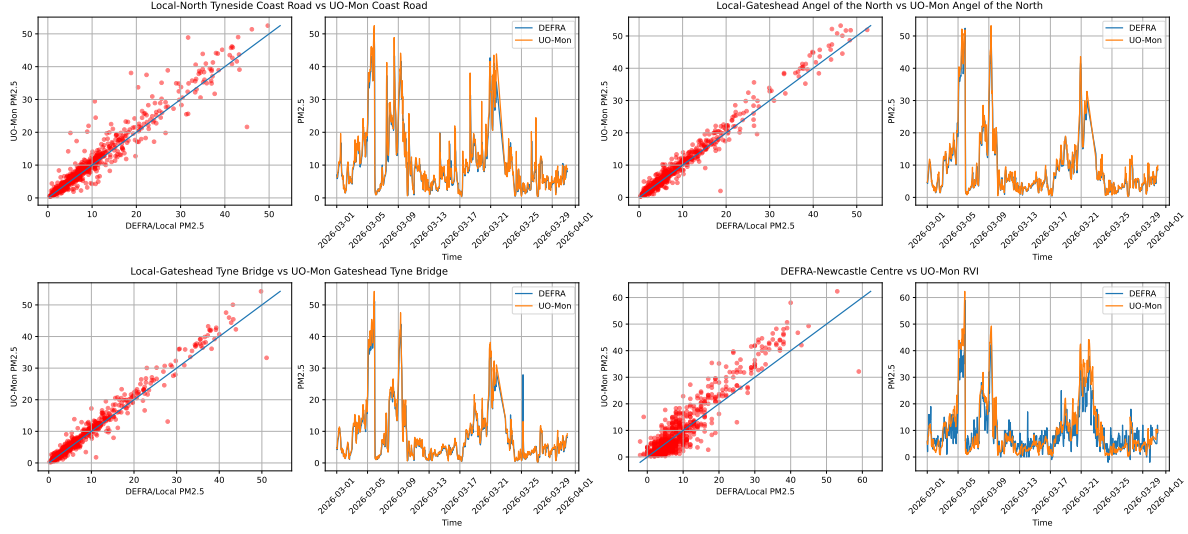


Figure 3: Comparison of PM2.5 concentrations measured by each DEFRA/Local sensor and its closest UO-Mon sensor over the study period. Only the closest UO-Mon sensor is included for each DEFRA/Local sensor.

Checking the nonlinear relationships seen in Figure 4, we examined in the agreement between Local-Gateshead A1 Dunston and Local-Gateshead Tyne Bridge since Figure 4 compares with UO-Mon Geateshead Tyne Bridge (in the same location) so we can understand if the disagreement is indeed related to distance and excluding sensor related issues. We should see a similar nonlinear relationship occuring as in Figure 5.

DEFRA/Local	UO-Mon	Distance km	Bird Flight Distance	Pearsons corr	Spearman's corr
Local-Gateshead Angel of the North	UO-Mon Angel of the North	0.003	0	0.986	0.962
Local-Gateshead Tyne Bridge	UO-Mon Gateshead Tyne Bridge	0.006	0	0.983	0.971

DEFRA/Local	UO-Mon	Distance km	Bird Flight Distance	Pearsons corr	Spearman's corr
DEFRA-Newcastle Cradlewell Roadside	UO-Mon RVI	1.643	0.023	0.974	0.95
Local-North Tyneside Coast Road	UO-Mon Coast Road	0.002	0	0.961	0.954
DEFRA-Newcastle Centre	UO-Mon RVI	0.468	0.007	0.926	0.812
Local-Gateshead A1 Dunston	UO-Mon Gateshead	3.37	0.049	0.846	0.834
Local-Gateshead Lychgate Court	UO-Mon Gateshead Tyne Bridge	0.572	0.007	0.741	0.789

Table 1: Distances between sensors in km and bird’s flight distance, Pearson’s and Spearman’s correlations for UO-Mon and closest DEFRA/Local sensors PM2.5 concentrations over long time interval.

4 Weekly data

We took the last week (30th March-5th April) of March 2026 to look at correlation comparisons in PM2.5 concentrations between the UO-Mesh sensors to UO-Mon sensors in Section 4.1. We also looked at correlational comparisons of PM2.5 and NOx/NO2 concentrations in the same sensors for all DEFRA, Local and UO-Mon sensors in Section 4.2.

4.1 PM2.5 comparisons between close sensors

Figure 7 shows closest UO-Mon sensors to UO-Mesh sensors scatter plots. Unfortunately, only two of the UO-Mon sensors have decent agreement with the two close UO-Mesh sensors, specifically UO-Mesh West Denton and Coast Road (next to the UO-Mon Coast Road sensor). Most others show little agreement however moderate positive correlation, UO-Mesh Heaton Road and Grainger Market have particularly high correlations for both Pearson’s and Spearman’s (seen in Table 2). Oddly, the UO-Mesh West Denton

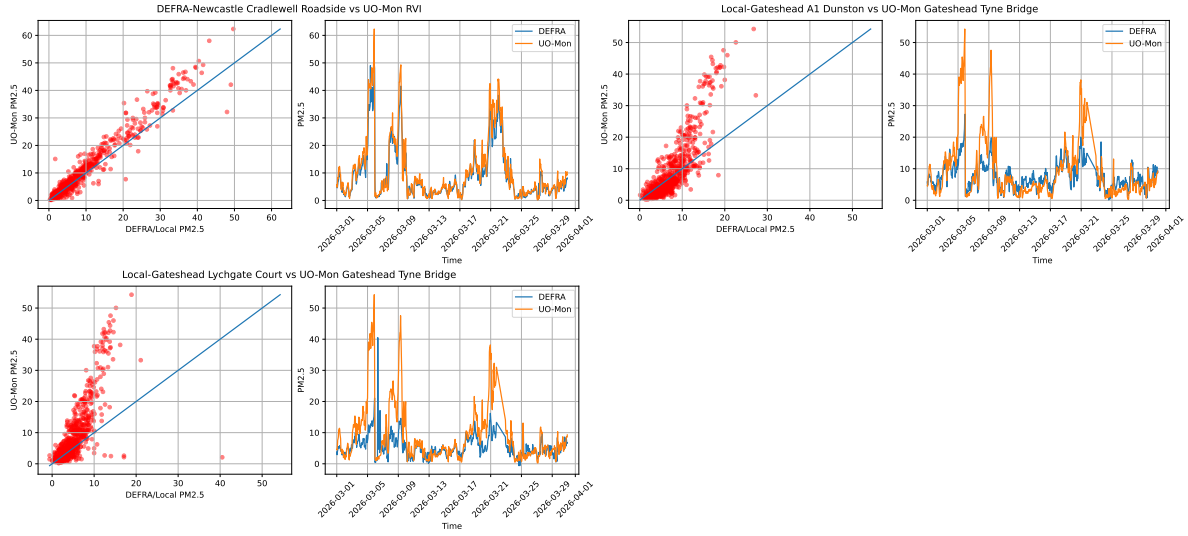


Figure 4: Comparison of PM2.5 concentrations measured by DEFRA/Local sensors and additional nearby UO-Mon sensors over the study period. These comparisons exclude the nearest UO-Mon sensor shown in Figure 3 and illustrate relationships with other matched UO-Mon sensors.

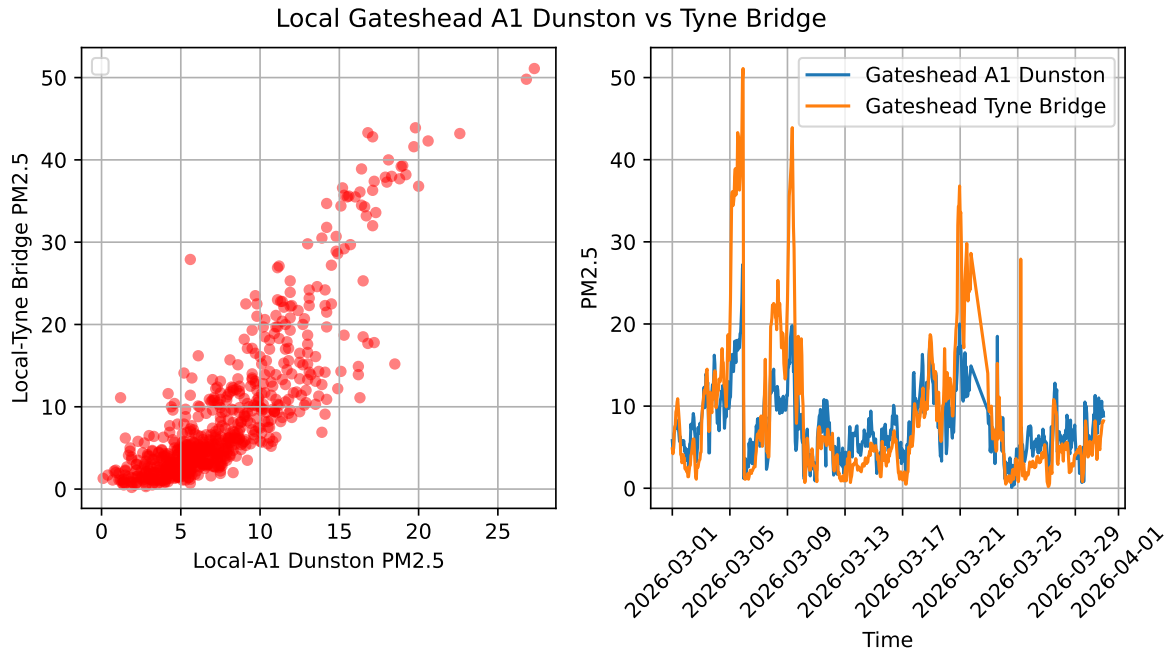


Figure 5: Comparison of PM2.5 concentrations measured by Local-Gateshead A1 Dunston and Local-Gateshead Tyne Bridge.

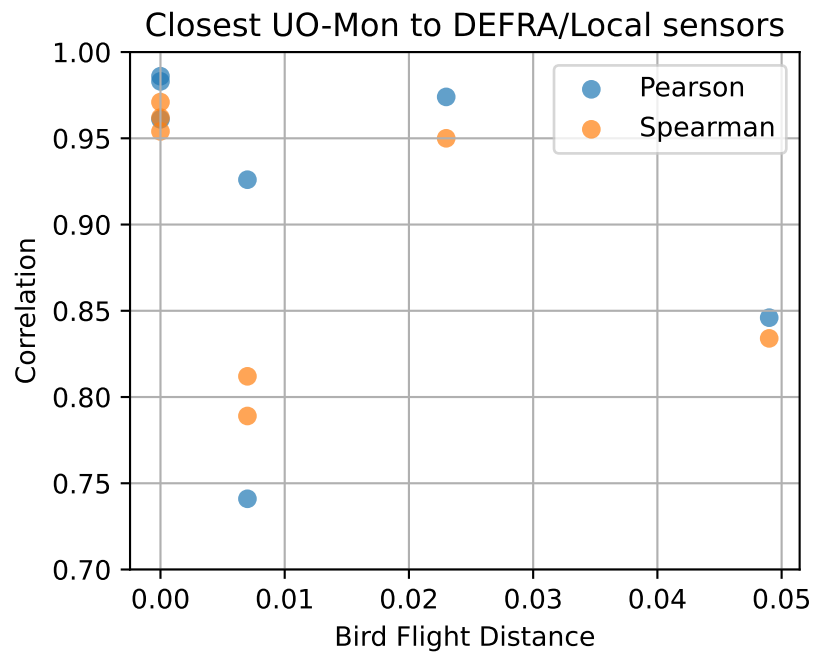


Figure 6: Correlation coefficients of each sensor pair from Figure 3 and Table 1 with their bird's flight distance.

sensor is much further away from the UO-Mon RVI sensor which is similarly compared to the UO-Mesh Grainger Market. Some sensors show little to no correlation however UO-Mesh Wingrove with UO-Mon RVI has very low Pearson's correlation and moderate Spearman's correlation, indicating some monotonic relation between then PM2.5 concentrations. There are some intriguing and hopeful structures in the data, these vastly differ week to week although the same correlation pattern between sensors is present which will need some further analysis.

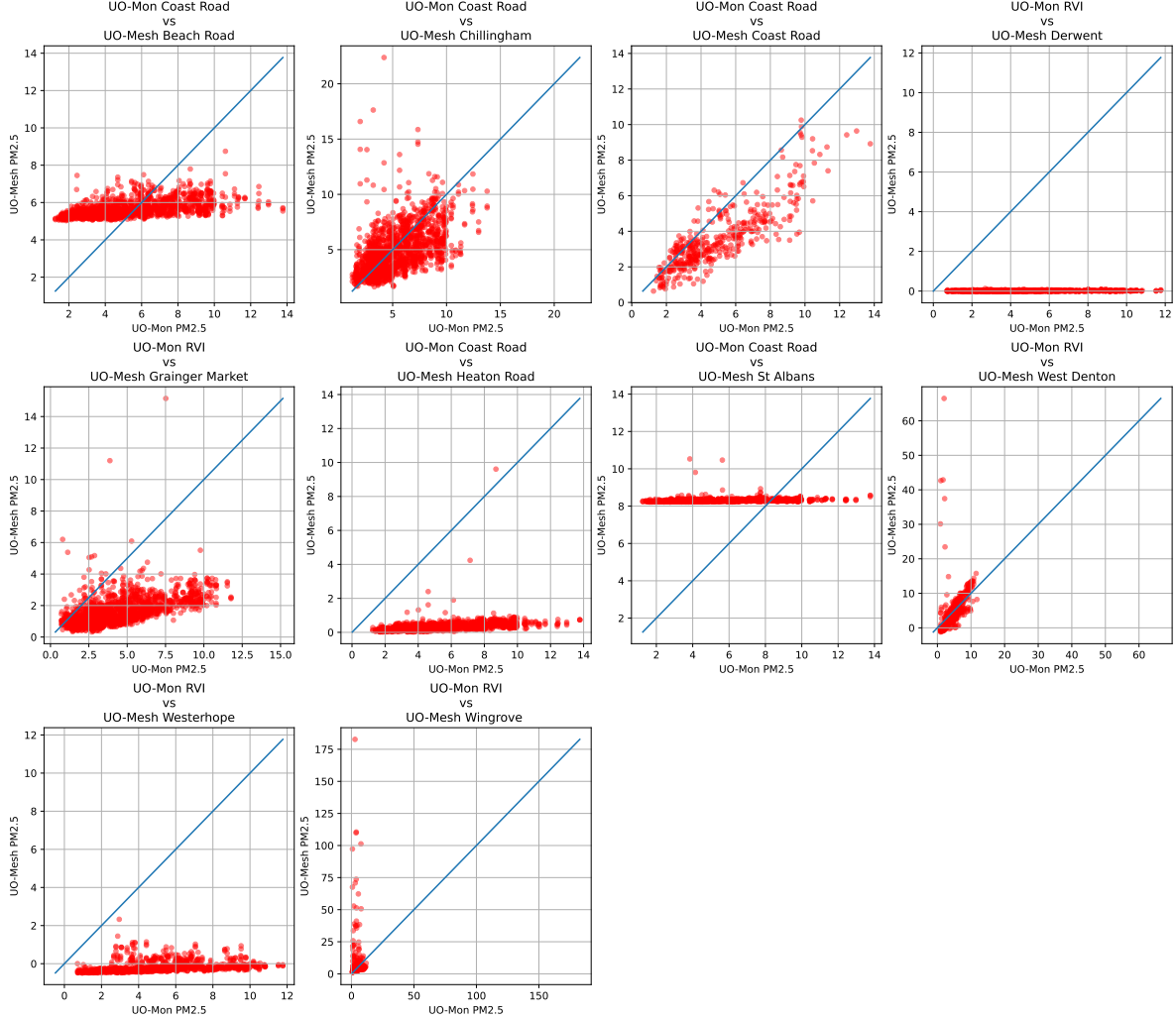


Figure 7: PM2.5 concentrations comparisons between UO-Mesh and closest UO-Mon sensors over short time interval.

UO-Mesh	UO-Mon	Distance km	Bird Flight Distance	Pearsons corr	Spearman's corr
UO-Mesh Coast Road	UO-Mon Coast Road	0.033	0	0.848	0.836
UO-Mesh Chillingham	UO-Mon Coast Road	1.757	0.021	0.544	0.615
UO-Mesh Grainger Market	UO-Mon RVI	0.754	0.007	0.538	0.64
UO-Mesh Beach Road	UO-Mon Coast Road	6.15	0.092	0.531	0.59
UO-Mesh West Denton	UO-Mon RVI	6.098	0.094	0.479	0.807
UO-Mesh Heaton Road	UO-Mon Coast Road	1.709	0.026	0.401	0.672
UO-Mesh Westerhope	UO-Mon RVI	6.045	0.089	0.385	0.691
UO-Mesh St Albans	UO-Mon Coast Road	1.904	0.018	0.231	0.499
UO-Mesh Derwent	UO-Mon RVI	5.252	0.076	0.151	0.165
UO-Mesh Wingrove	UO-Mon RVI	1.985	0.031	-0.014	0.519

Table 2: Distances between sensors in km and bird flight, Pearson’s and Spearman’s correlations for UO-Mesh and closest UO-Mon sensors PM2.5 concentrations over short time interval.

4.2 NOx and NO2 comparisons between same sensors

First NOx and NO2 concentration comparison in the DEFRA sensors to discern if PM2.5 has any traffic dependency in the same way that NO2 and NOx have. In Figure 8 scatter plots, we see very clearly here the systematic categorisation of the DEFRA Newcastle Centre sensor. Low correlation values can be seen for all in Table 3.

DEFRA	Pearsons corr NOx	Spearman's corr NOx	Pearsons corr NO2	Spearman's corr NO2
DEFRA-Newcastle Cradlewell Roadside	0.152	-0.014	0.142	0.1
DEFRA-Newcastle Centre	0.05	-0.016	0.093	-0.033

DEFRA	Pearsons corr NOx	Spearman's corr NOx	Pearsons corr NO2	Spearman's corr NO2
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Table 3: Pearson’s and Spearman’s correlations for DEFRA sensor PM2.5 concentrations and NOx, NO2 concentrations over short time interval.

In Figure 9, we can see scatter plots for NO2 comparison with PM2.5 concentration for the Local sensors as they not not collect NOx data. Moderate correlation values can be seen in Table 4 which suggest there may be some traffic dependence or possibly other factors like temperature.

	Pearsons corr NO2	Spearman's corr NO2
Local		
Local-North Tyneside Coast Road	0.439	0.371
Local-Gateshead Lychgate Court	0.339	0.398
Local-Gateshead Angel of the North	0.254	0.287
Local-Gateshead A1 Dunston	0.167	0.186
Local-Gateshead Tyne Bridge	0.120	0.129

Table 4: Pearson’s and Spearman’s correlations for Local sensor PM2.5 concentrations and NO2 concentrations over short time interval.

In Figure 10, scatter plots for NOx comparison with PM2.5 concentration for the UO-Mon sensors for now. NO2 concentration comparisons are still in process with some data download difficulties with UO-API. In Table 5, we see that some of the UO-Mon sensors show moderate positive correlation between PM2.5 and NOx, however the other half show little to no correlation.

	Pearsons corr NOx	Spearman's corr NOx
UO-Mon		
UO-Mon Coast Road	0.461	0.401
UO-Mon Angel of the North	0.231	0.199
UO-Mon Gateshead Tyne Bridge	0.107	0.118
UO-Mon RVI	0.079	0.134

Table 5: Pearson’s and Spearman’s correlations for UO-Mon sensor PM2.5 concentrations and NOx concentrations over short time interval.

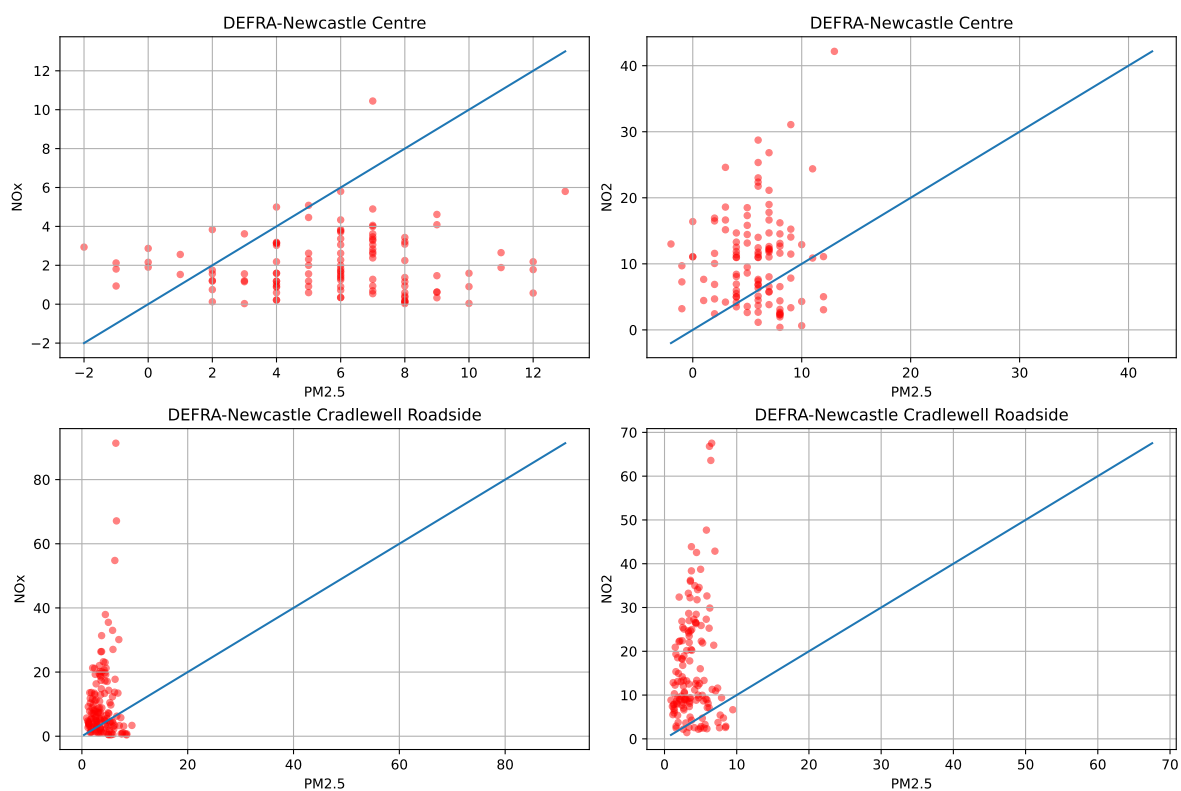


Figure 8: Comparison of PM2.5 and NOx and NO2 concentrations for DEFRA sensors over short time interval.

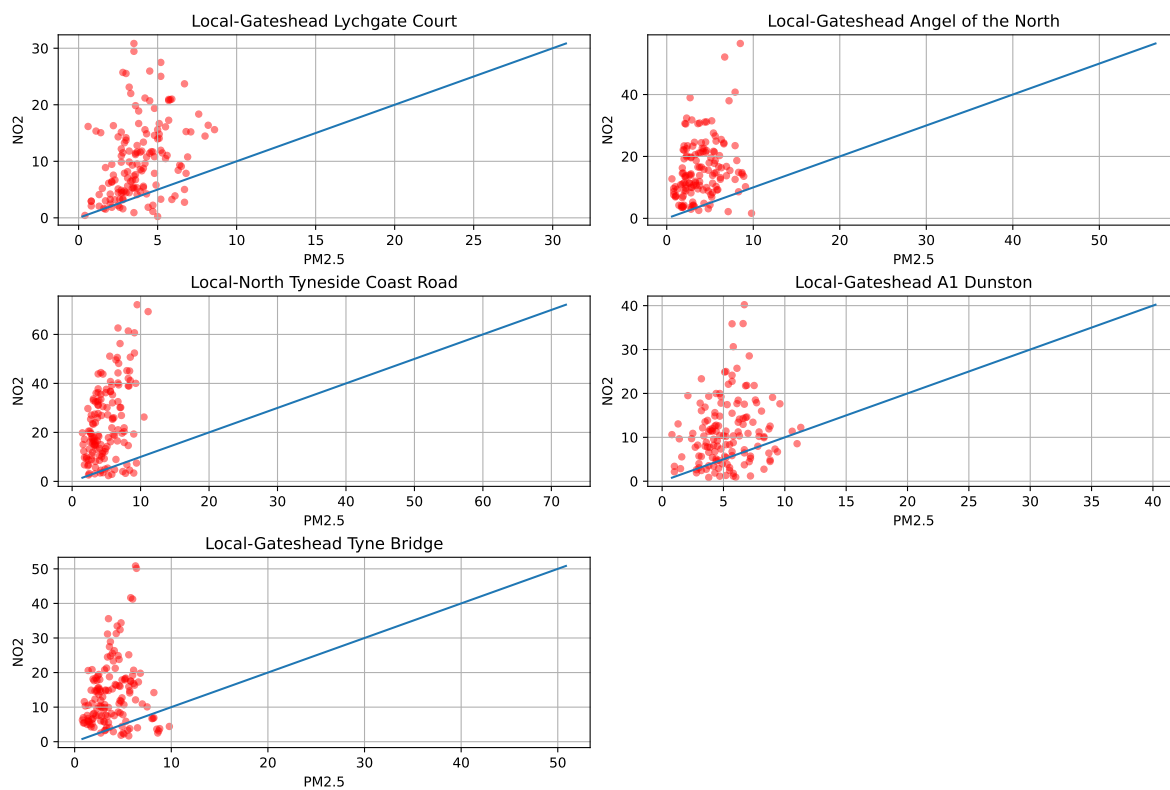


Figure 9: Comparisons of Local sensor PM2.5 and NO2 concentrations over short time interval.

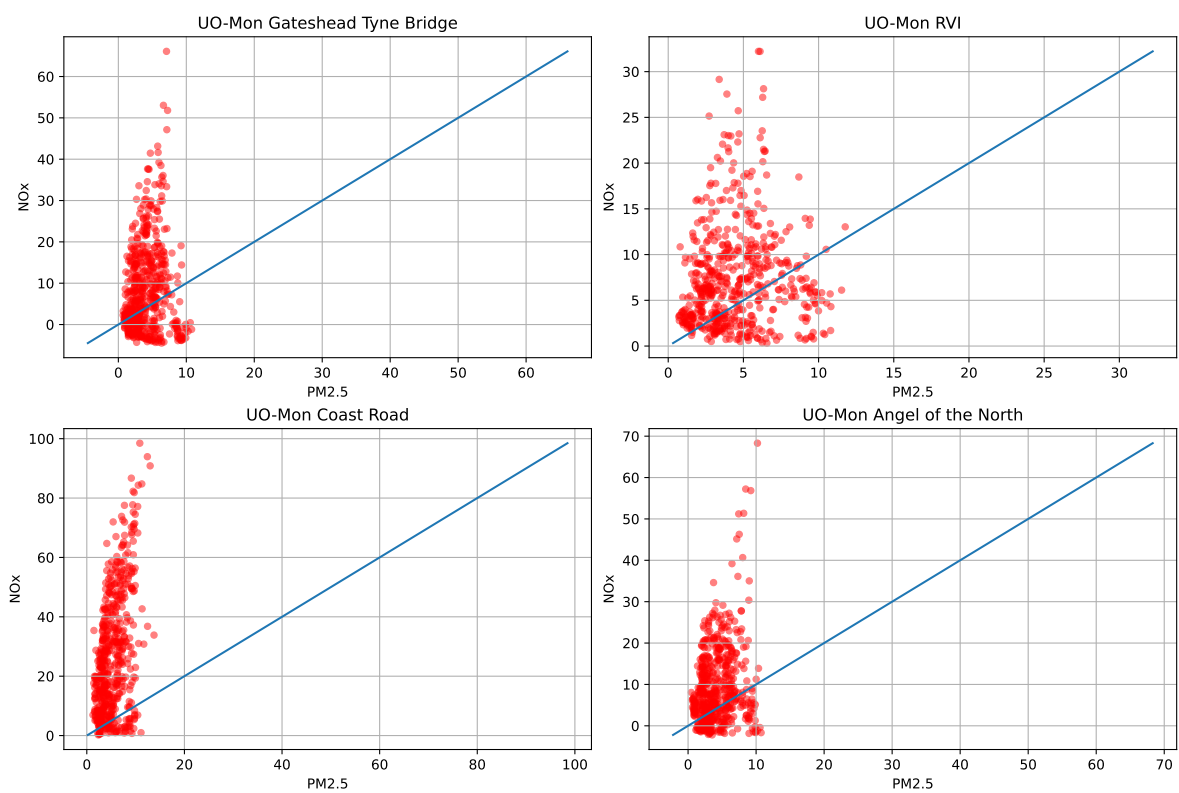


Figure 10: Comparisons of UO-Mon sensor PM2.5 and NOx concentrations over short time interval.

5 Attributes affecting PM2.5 concentraions

5.1 Wind direction and speed

Wind data from UO-Mon sensors was taken for the month of March 2026, which turned out to be faulty as seen from Figure 11 below showing constant speed and direction over the course of the entire day/week/month. Further to this we analysed wind data from two older UO sensors in Birtley and West Denton to see if we could dicern anything interesting.

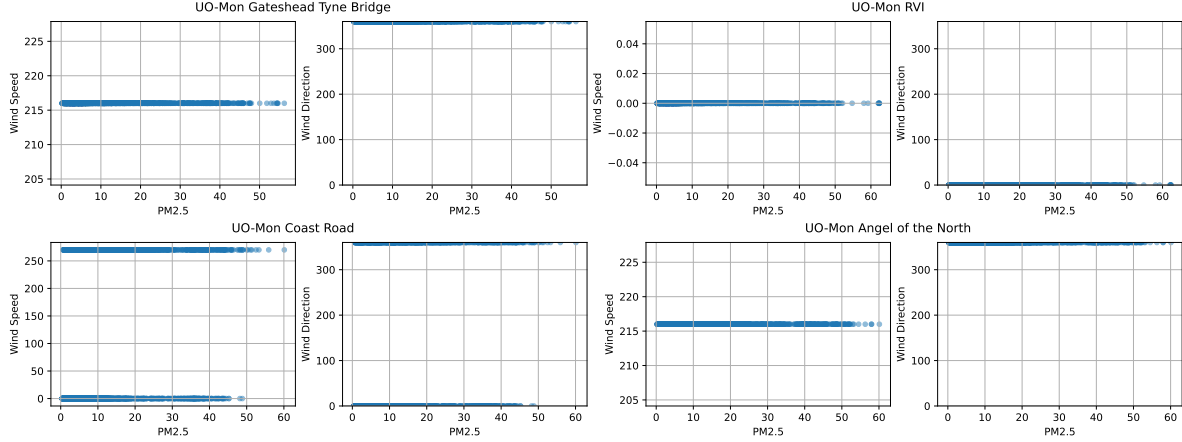


Figure 11: Wind speed and direction comparisons to PM2.5 concentrations from the UO-Mon sensors.

The two UO sensors in Birtley and West Denton seem to have data throughtout 2025, simiarly to previous we looked at March 2025 data, we analysed how PM2.5 concentrations behaved in the closest UO-Mon sensors and Local sensors which seems to show a relation in wind direction. Figure 12 shows maps of the closest UO-Mon/Local sensors to the Birtley and West Denton sensors. Figure 13 shows scatter plots of PM2.5 concentration to wind speed coloured by wind direction, rose plots of PM2.5 concentration to wind direction, and binned plots of wind speed and direction coloured by PM2.5 concentration for the closest UO-Mon sensors and Figure 14 shows the same for closes Local sensors. It is especially apparent in the rose plots that the higher PM2.5 concentrations are present when the wind direction in the first quarter (0-90 degrees) with some cross over into neighbouring quarters. This is less apparent in the Local sensor comparisons, although the trend is still present, which may be a result of PM2.5 data measured every hour in comparison to the UO-Mon sensors which measure every 15 minutes. From the scatter plot and binned plot we can also see the relation to wind speed where we tend to see consistently lower PM2.5 concentrations at higher wind speeds.

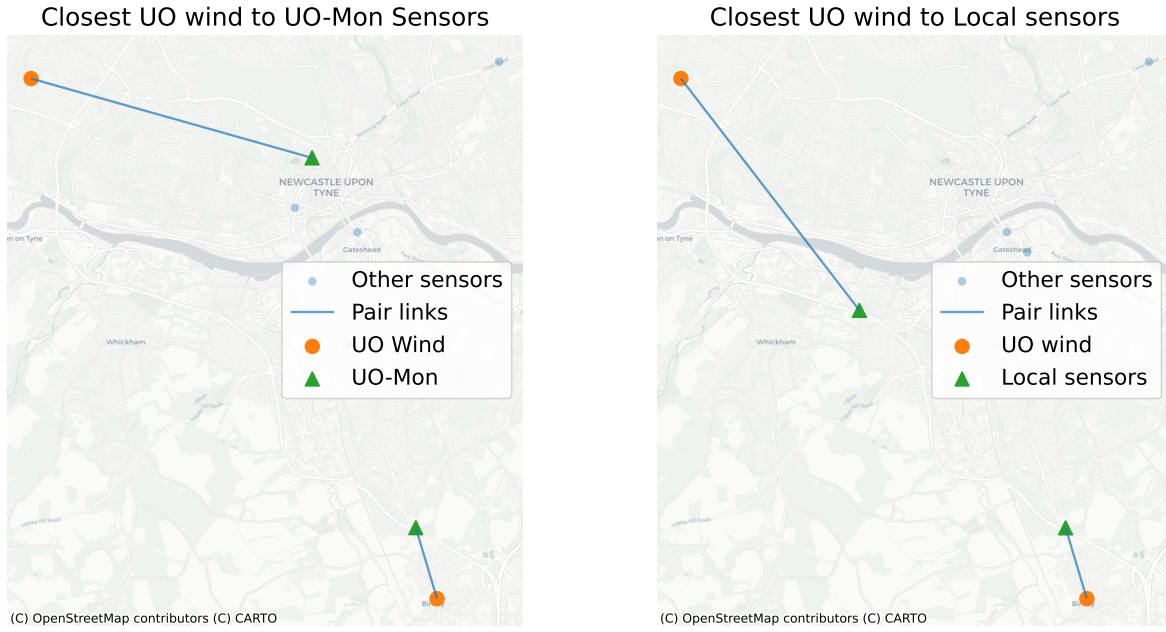


Figure 12: Maps of UO-Mesh sensors and Local sensors to UO wind sensors in Birtley and West Denton.

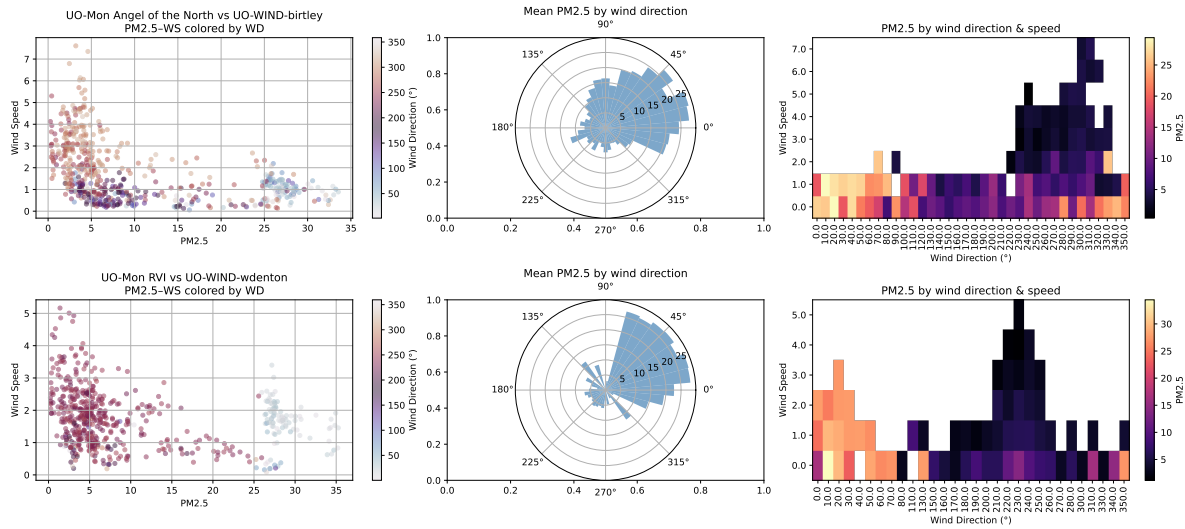


Figure 13: Scatter plots of PM2.5 concentration to wind speed coloured by wind direction, rose plots of PM2.5 concentration to wind direction, and binned plots of wind speed and direction coloured by PM2.5 concentration for the closest UO-Mon sensors to UO wind sensors in Birtley and West Denton.

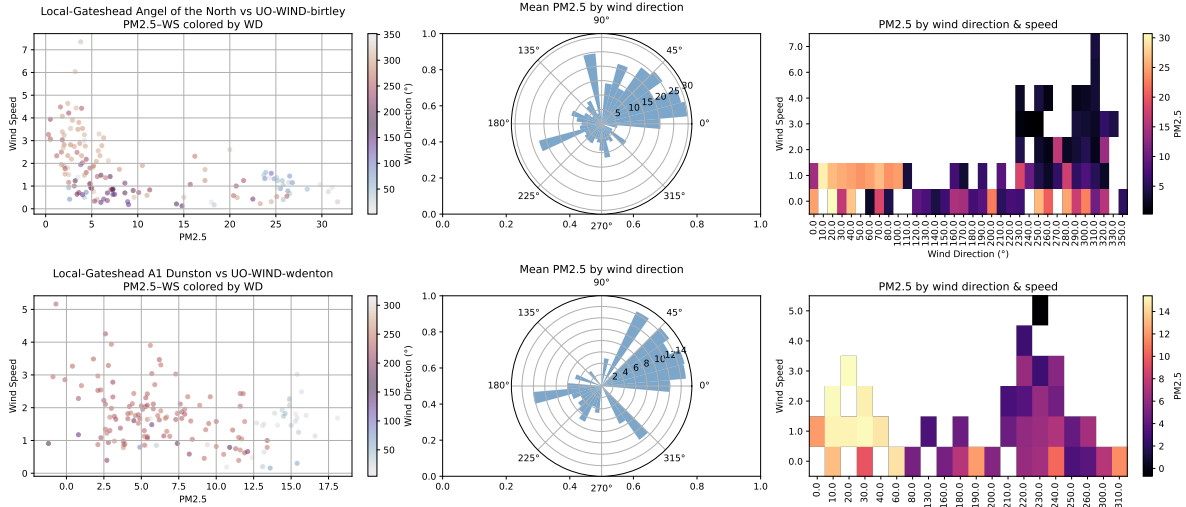


Figure 14: Scatter plots of PM2.5 concentration to wind speed coloured by wind direction, rose plots of PM2.5 concentration to wind direction, and binned plots of wind speed and direction coloured by PM2.5 concentration for the closest Local sensors to UO wind sensors in Birtley and West Denton.

5.2 Altitude

We have used Newcastle upon Tyne extent of the DEFRA/Local/UO-Mon sensor area Ordinate Survey digital elevation map to evaluate whether any concentration level could be attributed to altitude of the sensors. From Figure 15 where we look at mean and standard deviations for each sensor PM2.5 data over the course of the month with their altitude we see that there is a small positive correlation for mean and a moderate positive correlation for the standard deviation which indicates the higher altitude could affect higher variability in PM2.5 concentrations. We also examined at the time series of PM2.5 concentrations over the course of a week to understand if altitude has an effect on concentration levels temporally as seen in Figure 16 for which there are little obvious differing features.

5.3 Temperature

We compared UO-Mon sensor PM2.5 with the same sensor temperature data over March 2026 in order to discern if there is relationship as our analysis with the NOx and NO2 concentrations could suggest that other factors may be of interest. In Figure 17, we can see scatter plots for temperature comparison with PM2.5 concentration for the UO-Mon sensors which, alongside low correlation values seen in Table 6, suggest very little relation between temperature and PM2.5 concentrations for at least a monthly time scale.

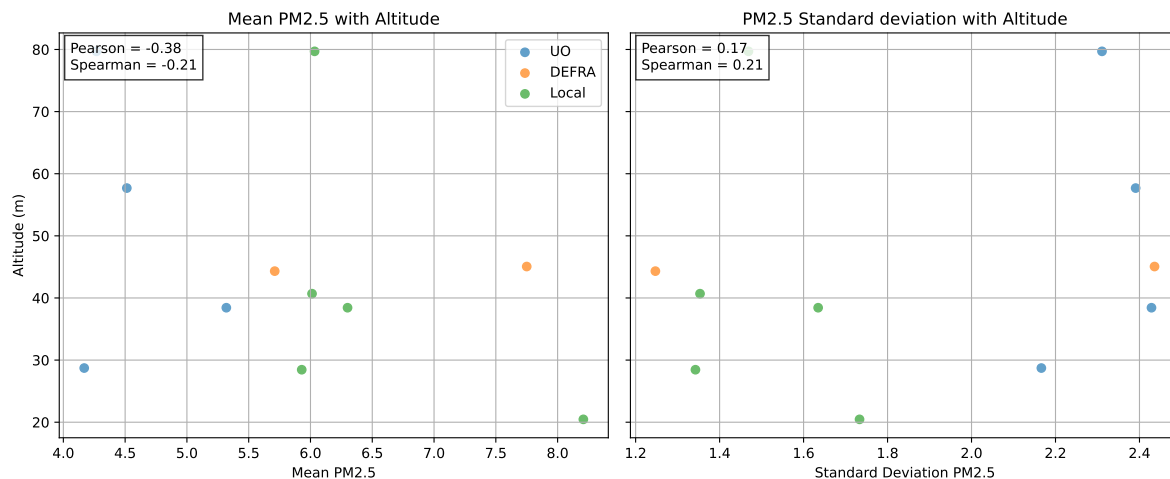


Figure 15: Altitude of DEFRA/Local/UO-Mon sensors with mean and standard deviation in PM2.5 concentration over long time interval.

	Pearsons corr	Spearman's corr
UO-Mon		
UO-Mon Coast Road	0.140	0.177
UO-Mon Angel of the North	0.114	0.082
UO-Mon RVI	0.086	0.073
UO-Mon Gateshead Tyne Bridge	0.061	0.039

Table 6: Pearson's and Spearman's correlations for UO-Mon sensor PM2.5 concentrations and temperature over long time interval.

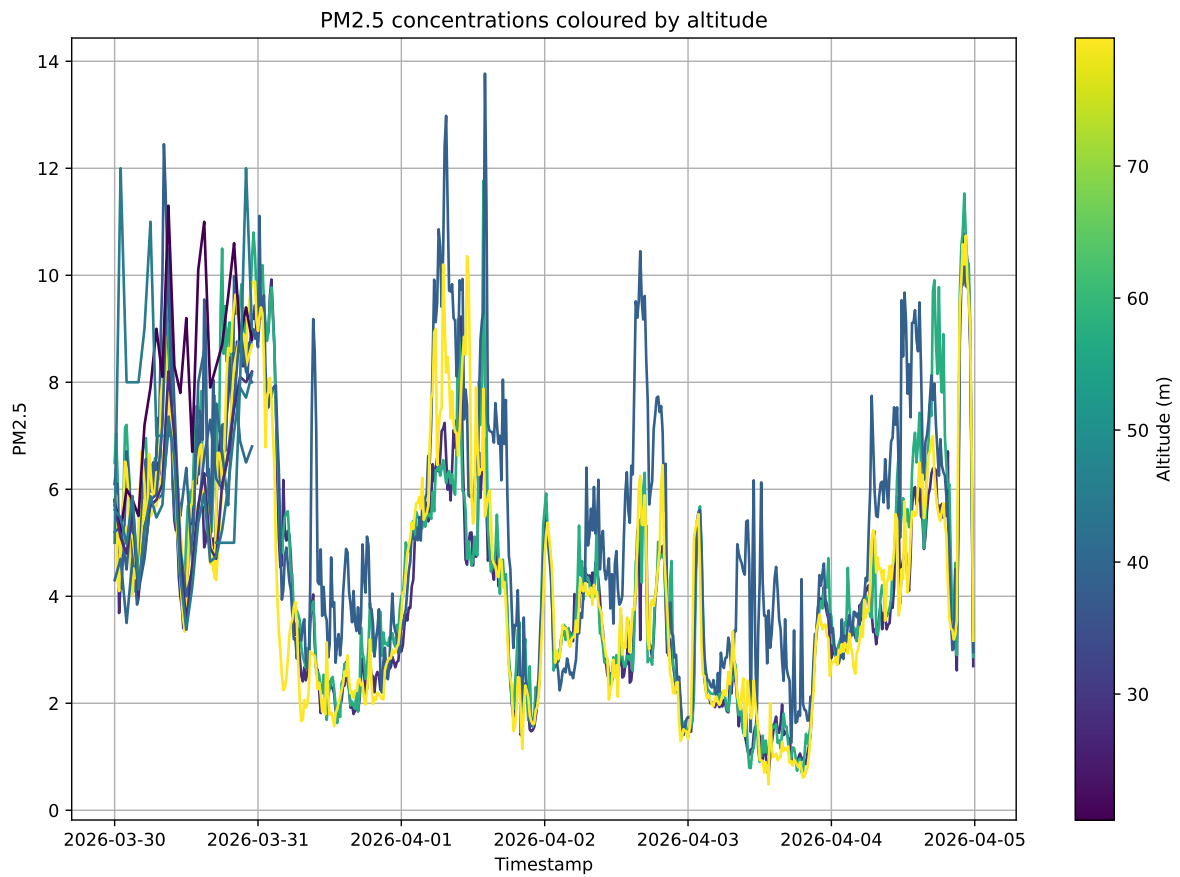


Figure 16: Timeseries of PM2.5 concentrations for different sensors coloured by altitude.

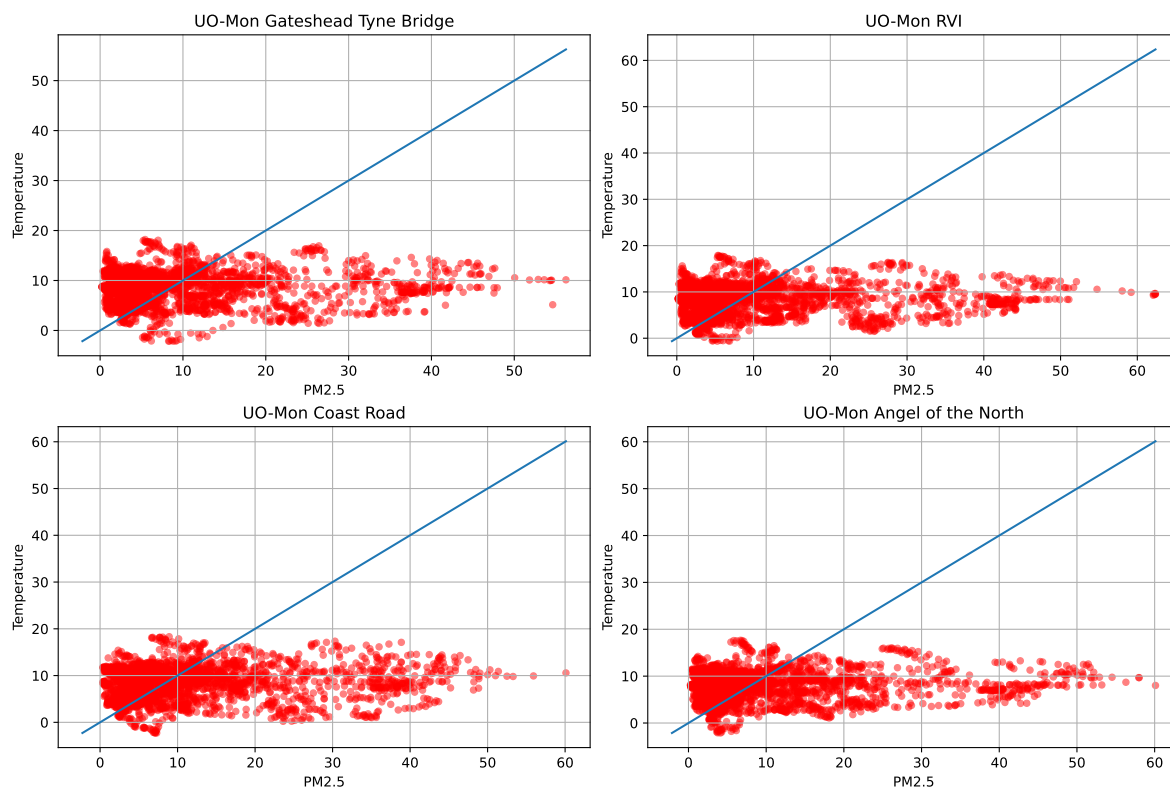


Figure 17: Comparison on PM2.5 concentration and temperature from UO-Mon sensors.